

Fig. 4: Actual-size PCB

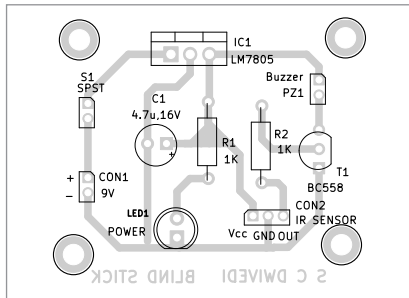


Fig. 5: Components side

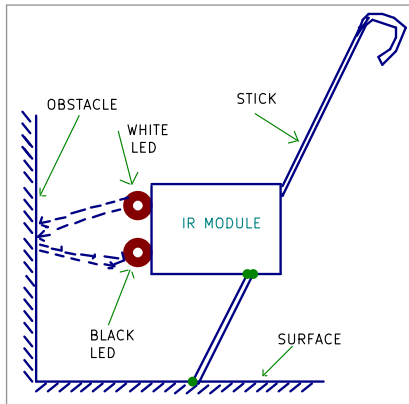


Fig. 6: Stick fitted with IR module

detected. Fig. 3 shows an image of the IR sensor module.

The working of the circuit is quite simple. When the lower section of the stick (where the sensor is fitted) approaches an obstacle, IR rays are reflected, causing the IR module to generate a low signal at its data pin. This activates transistor T1, triggering the buzzer to sound and alerting the user to an obstacle ahead, prompting the person to change direction.

In summary, the circuit instantly detects obstacles via the IR module, and the user receives an audible alert from the buzzer. This helps a user to avoid obstacles safely without collisions.

Construction and testing

A full-size, single-sided PCB layout for the blind stick circuit is shown in Fig. 4, with the component layout in Fig. 5. After assembling the circuit on the PCB, enclose it in a suitable cabinet with a window for the IR sensor module.

Alternatively, the circuit can be assembled on a general-purpose PCB and enclosed in a plastic box, ensuring that both the IR transmitter and receiver LEDs are positioned outside the box.

Installation steps

Installing a PCB on a walking stick for the visually impaired with an IR sensor is a precise task, requiring a step-by-step approach. Before installation, refer to Fig. 6.

Take the assembled PCB.

Identify a position near the bottom of the stick where the sensor can detect obstacles.

Fix the IR module with its enclosure onto the stick as shown in Fig. 6.

Secure the enclosure to the walking stick using screws.

Position the IR sensor facing forward or slightly downward to detect obstacles ahead.

Use insulated wires to route the power source securely along the stick. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-how-to-make-stick-for-the-blind>

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Anyone who stops learning is old, whether at twenty or eighty.

—Henry Ford

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